Package 'LHmom'

June 25, 2026

Title What the Package Does (One Line, Title Case)

Version 0.0.0.9000

Description What the package does (one paragraph).

License GPL (>= 3)

Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.3

Contents

bangkok1	*Annual Maximum Daily Rainfall for Bangkok, Thailand*

Description

A dataset containing the annual maximum series (AMS) of daily rainfall measurements in Bangkok, Thailand, from 1951 to 2025. This dataset is typically used for extreme value analysis and hydrological frequency modeling.

Usage

```
data(bangkok1)
```

Format

A data frame with 75 rows and 2 variables:

year The observation year (1951-2025).

rainfall The annual maximum daily rainfall in millimeters (mm).

Source

Thai Meteorological Department (TMD)

Examples

```
data(bangkok1)
head(bangkok1)
```

khonkaen	*Annual Maximum Daily Rainfall for Khon Kaen, Thailand*

Description

A dataset containing the annual maximum series (AMS) of daily rainfall measurements in Khon Kaen, Thailand, from 1984 to 2025.

Usage

```
data(khonkaen)
```

Format

A data frame with 42 rows and 2 variables:

year The observation year (1984-2025).

rainfall The annual maximum daily rainfall in millimeters (mm).

Source

Thai Meteorological Department (TMD)

Examples

```
data(khonkaen)
head(khonkaen)
```

`lh.pargev` *Estimate Parameters of the Generalized Extreme Value (GEV) Distribution using LH-moments*

Description

This function estimates the parameters of the Generalized Extreme Value (GEV) distribution based on the sample LH-moments. The estimation methodology follows Wang (1997). It provides two approaches for estimating the shape parameter: using Wang's predefined polynomial approximations or utilizing numerical optimization.

Usage

```
lh.pargev(data, eta = 1, opt = FALSE, ntry = 5)
```

Arguments

`data`	A numeric vector of data values.
`eta`	A non-negative integer (between 0 and 4) representing the order of the LH-moments. Default is 1.
`opt`	A logical value indicating the estimation method for the shape parameter. If `FALSE` (default), it uses Wang's polynomial approximation coefficients. If `TRUE`, it uses numerical optimization via `nleqslv`.
`ntry`	An integer specifying the maximum number of initialization attempts for the numerical optimization solver when `opt` = `TRUE`. Default is 5.

Value

A list containing:

- `type`: The distribution type (`"gev"`).
- `para`: A named numeric vector containing the estimated parameters (`xi` for location, `alpha` for scale, `k` for shape).
- `eta`: The order of the LH-moments used.
- `source`: The name of the function (`"lh.pargev"`).
- `ifail`: A numeric indicator of the optimization solver's status (0 for success, 5 for failure to converge).

References

Wang, Q. J. (1997). Using higher order L-moments for regional flood frequency analysis. *Water Resources Research*, 33(12), 2841-2848.

`lh.parggd` *Estimate Parameters of the Generalized Gumbel (GGD) Distribution using LH-moments*

Description

This function estimates the parameters of the Generalized Gumbel (GGD) distribution based on the sample LH-moments. It evaluates the GGD as a special case of the three-parameter Kappa distribution where the shape parameter `k` is fixed at 0.

Usage

```
lh.parggd(data, eta = 1)
```

Arguments

`data`	A numeric vector of data values.
`eta`	A non-negative integer (between 0 and 4) representing the order of the LH-moments. Default is 1.

Value

A list containing:

- `para`: A named numeric vector containing the estimated parameters (`mu` for location, `sigma` for scale, and `h` for shape).
- `type`: The distribution type (`"ggd"`).
- `source`: The name of the function (`"lh.parggd"`).
- `eta`: The order of the LH-moments used.

`lh.parglo` *Estimate Parameters of the Generalized Logistic (GLO) Distribution using LH-moments*

Description

This function estimates the parameters of the Generalized Logistic (GLO) distribution based on the sample LH-moments. The analytical formulas for the parameter estimation are derived based on the methodology presented by Meshgi and Khalili (2009).

Usage

```
lh.parglo(data, eta = 1)
```

Arguments

`data`	A numeric vector of data values.
`eta`	A non-negative integer (between 0 and 4) representing the order of the LH-moments. Default is 1.

Value

A list containing:

- `type`: The distribution type (`"glo"`).
- `para`: A named numeric vector containing the estimated parameters (`mu` for location, `sigma` for scale, `k` for shape).
- `eta`: The order of the LH-moments used.
- `source`: The name of the function (`"lh.parglo"`).
- `ifail`: A numeric indicator of success (0 for success).

References

Meshgi, A., & Khalili, D. (2009). Comprehensive evaluation of regional flood frequency analysis by L- and LH-moments. *Stochastic Environmental Research and Risk Assessment (SERRA)*, 23(1), 137-152.

`lh.pargno` *Estimate Parameters of the Generalized Normal (GNO) Distribution using LH-moments*

Description

This function estimates the parameters of the Generalized Normal (GNO) distribution based on the sample LH-moments. It provides two methods for estimating the shape parameter: using predefined polynomial approximations or utilizing numerical optimization.

Usage

```
lh.pargno(data, eta = 1, opt = FALSE, ntry = 5)
```

Arguments

`data`	A numeric vector of data values.
`eta`	A non-negative integer (between 0 and 4) representing the order of the LH-moments. Default is 1.
`opt`	A logical value indicating the estimation method for the shape parameter. If `FALSE` (default), it uses a polynomial approximation. If `TRUE`, it uses numerical optimization via `nleqslv`.
`ntry`	An integer specifying the maximum number of initialization attempts for the numerical optimization solver when `opt = TRUE`. Default is 5.

Value

A list containing:

- `type`: The distribution type (`"gno"`).
- `para`: A named numeric vector containing the estimated parameters (`xi` for location, `alpha` for scale, `k` for shape).
- `eta`: The order of the LH-moments used.

- `source`: The name of the function ("`lh.pargno`").
- `ifail`: A numeric indicator of the optimization solver's status (0 for success, 5 for failure to converge).

`lh.pargpa` *Estimate Parameters of the Generalized Pareto (GPA) Distribution using LH-moments*

Description

This function estimates the parameters of the Generalized Pareto (GPA) distribution based on the sample LH-moments. The analytical formulas for the parameter estimation are derived based on the methodology presented by Meshgi and Khalili (2009).

Usage

```
lh.pargpa(data, eta = 1)
```

Arguments

`data`	A numeric vector of data values.
`eta`	A non-negative integer (between 0 and 4) representing the order of the LH-moments. Default is 1.

Value

A list containing:

- `type`: The distribution type ("`gpa`").
- `para`: A named numeric vector containing the estimated parameters (`mu` for location, `sigma` for scale, `k` for shape).
- `eta`: The order of the LH-moments used.
- `source`: The name of the function ("`lh.pargpa`").
- `ifail`: A numeric indicator of success (0 for success).

References

Meshgi, A., & Khalili, D. (2009). Comprehensive evaluation of regional flood frequency analysis by L- and LH-moments. *Stochastic Environmental Research and Risk Assessment (SERRA)*, 23(1), 137-152.

lh.pargum — *Estimate Parameters of the Gumbel Distribution using LH-moments*

Description

This function estimates the parameters of the Gumbel distribution based on the sample LH-moments. The Gumbel distribution is treated as a special case of the Generalized Extreme Value (GEV) distribution with the shape parameter (k) fixed at 0. The analytical estimation utilizes the first two LH-moments and the Euler-Mascheroni constant.

Usage

```
lh.pargum(data, eta = 1)
```

Arguments

`data`	A numeric vector of data values.
`eta`	A non-negative integer (between 0 and 4) representing the order of the LH-moments. Default is 1.

Value

A list containing:

- `type`: The overarching distribution family (`"gev"`).
- `para`: A named numeric vector containing the estimated parameters (`xi` for location, `alpha` for scale, and `k` = 0 for shape).
- `eta`: The order of the LH-moments used.
- `source`: The name of the function (`"lh.pargum"`).
- `model`: The specific model name (`"gum"`).

lh.park3d.hfix — *Estimate Parameters of the Three-Parameter Kappa Distribution with Fixed h*

Description

This function estimates the parameters of the three-parameter Kappa distribution based on the sample LH-moments, given a fixed value for the shape parameter `h` in the four-parameter Kappa distribution. The function utilizes numerical optimization (`nleqslv`) to solve for the shape parameter `k`. If the numerical solver fails to converge, the function implements a fallback mechanism by adopting the `k` parameter estimated from the Generalized Extreme Value (GEV) distribution.

Usage

```
lh.park3d.hfix(data, eta = 1, hfix = 0, ntry = 10)
```

Arguments

`data`	A numeric vector of data values.
`eta`	A non-negative integer (between 0 and 4) representing the order of the LH-moments. Default is 1.
`hfix`	A numeric scalar representing the fixed value for the shape parameter `h`. Default is 0.
`ntry`	An integer specifying the maximum number of initialization attempts for the numerical optimization solver. Default is 10.

Value

A list containing:

- `type`: The distribution type (`"kap"`).
- `para`: A named numeric vector containing the estimated parameters (`mu` for location, `sigma` for scale, `k` for shape, and `hfix`).
- `eta`: The order of the LH-moments used.
- `source`: The name of the function (`"lh.park3d.hfix"`).
- `ifail`: A numeric indicator of the optimization solver's status (0 for success, 2 for partial success/limit reached, 5 for failure).
- `hfix`: The fixed value of the shape parameter `h` used in the estimation.
- `precision`: The final function value (`fvec`) from the solver, indicating the precision of the root found.

`lh.park3d.kfix` *Estimate Parameters of the Three-Parameter Kappa Distribution with Fixed k*

Description

This function estimates the parameters of the three-parameter Kappa distribution based on the sample LH-moments, given a fixed value for the shape parameter `k` in the four-parameter Kappa distribution. It utilizes numerical optimization (`nleqslv`) to solve for the second shape parameter `h`. If the numerical solver fails to converge, the function implements a fallback mechanism by adopting the `h` parameter estimated from the four-parameter Kappa distribution (`lh.parkap`).

Usage

```
lh.park3d.kfix(data, eta = 1, kfix = 0, hlow = NULL, ntry = 10)
```

Arguments

`data`	A numeric vector of data values.
`eta`	A non-negative integer (between 0 and 4) representing the order of the LH-moments. Default is 1.
`kfix`	A numeric scalar representing the fixed value for the shape parameter `k`. Default is 0.
`hlow`	A numeric scalar representing the lower bound for the shape parameter `h`. If `NULL`, it defaults to `-eta - 1`.
`ntry`	An integer specifying the maximum number of initialization attempts for the numerical optimization solver. Default is 10.

Value

A list containing:

- `type`: The distribution type ("kap").
- `para`: A named numeric vector containing the estimated parameters (`mu` for location, `sigma` for scale, `kfix`, and `h` for shape).
- `eta`: The order of the LH-moments used.
- `source`: The name of the function ("lh.park3d.kfix").
- `ifail`: A numeric indicator of the optimization solver's status (0 for success, 2 for partial success/limit reached, 5 for failure).
- `kfix`: The fixed value of the shape parameter `k` used in the estimation.
- `precision`: The final function value (`fvec`) from the solver, indicating the precision of the root found.

`lh.parkap` *Estimate Parameters of the Four-Parameter Kappa Distribution using LH-moments*

Description

This function estimates the parameters of the four-parameter Kappa (K4D) distribution based on the sample LH-moments. The estimation methodology follows Murshed et al. (2014). It utilizes numerical optimization (`nleqslv`) to simultaneously solve for the two shape parameters, `k` and `h`. If the optimization fails or becomes unstable, the function provides fallback mechanisms utilizing fixed-parameter Kappa estimations. When `eta = 0`, it defaults to the ordinary L-moments estimation via `lmomco::parkap`.

Usage

```
lh.parkap(
  data,
  eta = 1,
  snap.tau4 = TRUE,
  nudge.tau4 = 1e-05,
  hlow = NULL,
  ntry = 10
)
```

Arguments

`data`	A numeric vector of data values.
`eta`	A non-negative integer (between 0 and 4) representing the order of the LH-moments. Default is 1.
`snap.tau4`	A logical value indicating whether to snap the sample L-kurtosis (`tau4`) downward if it slightly exceeds the theoretical upper bound. Default is `TRUE`.
`nudge.tau4`	A small numeric value used to adjust `tau4` downward if `snap.tau4 = TRUE`. Default is `1e-5`.
`hlow`	A numeric scalar representing the lower bound for the shape parameter `h`. If `NULL`, it defaults to `-eta - 1`.
`ntry`	An integer specifying the maximum number of initialization attempts for the numerical optimization solver. Default is 10.

Value

A list containing:

- `type`: The distribution type ("`kap`").
- `para`: A named numeric vector containing the estimated parameters (`mu` for location, `sigma` for scale, `k` and `h` for shapes).
- `eta`: The order of the LH-moments used.
- `source`: The name of the function ("`lh.parkap`").
- `ifail`: A numeric indicator of the optimization solver's status (0 for success, 1 for fallback success, 5 for failure).
- `precision`: The final function values from the solver.
- `ifailtext`: A descriptive message regarding the estimation success or failure.

References

Murshed, S., Seo, Y.A., Park, J.S. (2014). LH-moment estimation of a four parameter kappa distribution with hydrologic applications. *Stochastic Environmental Research and Risk Assessment*, 28, 253-262.

`lh.parpe3` *Estimate Parameters of the Pearson Type III (PE3) Distribution using LH-moments*

Description

This function estimates the parameters of the Pearson Type III (PE3) distribution based on the sample LH-moments. It provides two estimation methods: using predefined polynomial coefficients (Log-Log Split Degree 5 based on sample LH-skewness) or numerical optimization via `nleqslv`. If the numerical solver fails to converge, the function falls back to the ordinary L-moments estimation (`eta = 0`).

Usage

```
lh.parpe3(data, eta = 1, opt = FALSE, ntry = 10)
```

Arguments

`data`	A numeric vector of data values.
`eta`	A non-negative integer (between 0 and 4) representing the order of the LH-moments. Default is 1.
`opt`	A logical value. If `FALSE` (default), it estimates parameters using polynomial approximations. If `TRUE`, it utilizes numerical optimization.
`ntry`	An integer specifying the maximum number of initialization attempts for the numerical optimization solver when `opt = TRUE`. Default is 10.

Value

A list containing:

- `type`: The distribution type (`"pe3"`).
- `para`: A named numeric vector containing the estimated parameters (`mu` for location, `sigma` for scale, `gamma` for shape).
- `eta`: The order of the LH-moments used.
- `source`: The name of the function (`"lh.parpe3"`).
- `ifail`: A numeric indicator of the optimization solver's status (0 for success, 2 for partial success, 5 for failure).
- `precision`: The final function value (`fvec`) from the solver.

`lh.qqplot` *Quantile-Quantile Plot for LH-moments Fitted Distributions*

Description

Generates a Quantile-Quantile (Q-Q) plot to visually assess the goodness-of-fit of a probability distribution fitted using higher-order L-moments (LH-moments). The function plots the empirical quantiles of the data against the theoretical quantiles of the specified distribution. It optionally calculates and displays bootstrap confidence intervals to evaluate model uncertainty.

Usage

```
lh.qqplot(
  data,
  fit_obj,
  main = "Q-Q Plot",
  ci = FALSE,
  ci.level = 0.95,
  n.boot = 500,
  ...
)
```

Arguments

`data`	A numeric vector of observations or a data frame with 1 column.
`fit_obj`	An object returned by any `lh.par*` function containing the fitted parameters.
`main`	Title of the plot.
`ci`	Logical; if `TRUE`, calculates and plots the bootstrap confidence interval.
`ci.level`	Numeric; confidence level for the interval (e.g., 0.95 for 95%).
`n.boot`	Number of bootstrap samples for CI (default is 500).
`...`	Additional graphical parameters passed to `plot()`.

Value

An invisible data frame containing theoretical quantiles, empirical quantiles, and CI bounds (if calculated).

`lhmom.gev` *Calculate Theoretical LH-moments for the Generalized Extreme Value (GEV) Distribution*

Description

This function computes the theoretical LH-moments and LH-moment ratios for the Generalized Extreme Value (GEV) distribution given its parameters. The computations are derived based on the formulas provided by Wang (1997).

Usage

```
lhmom.gev(para, eta = NULL)
```

Arguments

`para`	A numeric vector of three parameters: c(xi, alpha, k), corresponding to the location, scale, and shape parameters of the GEV distribution, respectively.
`eta`	A non-negative integer (between 0 and 4) representing the order of the LH-moments.

Value

A list containing:

- `lambdas`: A named numeric vector of the first four theoretical LH-moments.
- `ratios`: A named numeric vector of the corresponding LH-moment ratios.
- `eta`: The order of the LH-moments used.

References

Wang, Q. J. (1997). Using higher order L-moments for regional flood frequency analysis. *Water Resources Research*, 33(12), 2841-2848.

`lhmom.ggd` *Theoretical LH-moments for the Generalized Gumbel (GGD) Distribution*

Description

This function computes the theoretical LH-moments for the Generalized Gumbel (GGD) distribution. It is evaluated as a special case of the four-parameter Kappa distribution with the shape parameter k set to 0.

Usage

```
lhmom.ggd(para = NULL, eta = 1)
```

Arguments

`para`	A numeric vector of three parameters: c(mu, sigma, h).
`eta`	The order of LH-moments (default is 1).

Value

A list containing the calculated LH-moments and the distribution type ("ggd").

`lhmom.glo` *Theoretical LH-moments for the Generalized Logistic (GLO) Distribution*

Description

This function computes the theoretical LH-moments for the Generalized Logistic (GLO) distribution. It is evaluated as a special case of the four-parameter Kappa distribution with the shape parameter h set to -1.

Usage

```
lhmom.glo(para = NULL, eta = 1)
```

Arguments

`para`	A numeric vector of three parameters: c(mu, sigma, k).
`eta`	The order of LH-moments (default is 1).

Value

A list containing the calculated LH-moments and the distribution type ("glo").

`lhmom.gno` *Calculate Theoretical LH-moments for the Generalized Normal (GNO) Distribution*

Description

This function computes the theoretical LH-moments and LH-moment ratios for the Generalized Normal (GNO) distribution given its parameters. When the order `eta = 0`, it falls back to the ordinary L-moments using the `lmomco` package.

Usage

```
lhmom.gno(para = NULL, eta = 1)
```

Arguments

`para`	A numeric vector of three parameters: c(xi, alpha, k), corresponding to the location, scale, and shape parameters of the GNO distribution, respectively.
`eta`	A non-negative integer (between 0 and 4) representing the order of the LH-moments. Default is 1.

Value

A list containing:

- `lambdas`: A named numeric vector of the first four theoretical LH-moments.
- `ratios`: A named numeric vector of the corresponding LH-moment ratios.
- `eta`: The order of the LH-moments used.
- `type`: The distribution type (`"gno"`).

`lhmom.gpa` *Theoretical LH-moments for the Generalized Pareto (GPA) Distribution*

Description

This function computes the theoretical LH-moments for the Generalized Pareto (GPA) distribution. It is evaluated as a special case of the four-parameter Kappa distribution with the shape parameter h set to 1.

Usage

```
lhmom.gpa(para = NULL, eta = 1)
```

Arguments

`para`	A numeric vector of three parameters: c(mu, sigma, k).
`eta`	The order of LH-moments (default is 1).

Value

A list containing the calculated LH-moments and the distribution type ("gpa").

`lhmom.gum` *Theoretical LH-moments for Gumbel Distribution*

Description

This function computes the theoretical LH-moments for the Gumbel distribution by evaluating it as a special case of the Generalized Extreme Value (GEV) distribution with the shape parameter set to zero.

Usage

```
lhmom.gum(para = NULL, eta = 1)
```

Arguments

`para`	A vector of parameters c(mu, sigma)
`eta`	The order of LH-moments (default is 1).

Value

A list containing the LH-moments and distribution type.

`lhmom.k3d.hfix` *Theoretical LH-moments for the Three-Parameter Kappa Distribution with Fixed h*

Description

This function computes the theoretical LH-moments for the three-parameter Kappa distribution where the shape parameter h is fixed.

Usage

```
lhmom.k3d.hfix(para = NULL, eta = 1, hfix = 0)
```

Arguments

`para`	A numeric vector of three parameters: c(mu, sigma, k).
`eta`	The order of LH-moments (default is 1).
`hfix`	The fixed numeric value for the shape parameter h (default is 0).

Value

A list containing the calculated LH-moments and the distribution type ("kap").

`lhmom.k3d.kfix` *Theoretical LH-moments for the Three-Parameter Kappa Distribution with Fixed k*

Description

This function computes the theoretical LH-moments for the three-parameter Kappa distribution where the shape parameter k is fixed.

Usage

```
lhmom.k3d.kfix(para = NULL, eta = 1, kfix = 0)
```

Arguments

`para`	A numeric vector of three parameters: c(mu, sigma, h).
`eta`	The order of LH-moments (default is 1).
`kfix`	The fixed numeric value for the shape parameter k (default is 0).

Value

A list containing the calculated LH-moments and the distribution type ("kap").

lhmom.kap — *Calculate Theoretical LH-moments for the Four-Parameter Kappa Distribution*

Description

This function computes the theoretical LH-moments and LH-moment ratios for the four-parameter Kappa (K4D) distribution given its parameters. The computations are analytically derived based on the formulas presented by Murshed et al. (2014).

Usage

```
lhmom.kap(para = NULL, eta = 1)
```

Arguments

`para`	A numeric vector of four parameters: c(xi, alpha, k, h), corresponding to the location, scale, and the two shape parameters of the K4D distribution.
`eta`	A non-negative integer (between 0 and 4) representing the order of the LH-moments. Default is 1.

Value

A list containing:

- `lambdas`: A named numeric vector of the first four theoretical LH-moments.
- `ratios`: A named numeric vector of the corresponding LH-moment ratios.
- `eta`: The order of the LH-moments used.
- `type`: The distribution type (`"kap"`).

References

Murshed, S., Seo, Y.A., Park, J.S. (2014). LH-moment estimation of a four parameter kappa distribution with hydrologic applications. *Stochastic Environmental Research and Risk Assessment*, 28, 253-262.

lhmom.pe3 — *Calculate Theoretical LH-moments for the Pearson Type III (PE3) Distribution*

Description

This function computes the theoretical LH-moments and LH-moment ratios for the Pearson Type III (PE3) distribution. It supports parameter inputs either as a `vec2par` object from the `lmomco` package or as a standard numeric vector.

Usage

```
lhmom.pe3(para, eta = 0)
```

Arguments

`para`	A list containing a `vec2par` object (with `mu`, `sigma`, `gamma`) OR a numeric vector of length 3 `c(xi, alpha, beta)`.
`eta`	A non-negative integer representing the order of the LH-moments. Default is 0 (which corresponds to the standard L-moments).

Value

A list containing:

- `lambdas`: A named numeric vector of the first four theoretical LH-moments.
- `ratios`: A named numeric vector of the corresponding LH-moment ratios.
- `trim`: The trim level (fixed at 0 for LH-moments compatibility).
- `type`: The distribution type (`"pe3"`).
- `eta`: The order of the LH-moments used.
- `source`: The name of the function (`"lhmom.pe3"`).

`lhmoms` *Calculate Sample LH-moments*

Description

This function computes the sample LH-moments and LH-moment ratios from a numeric dataset. The estimation is based on the methodology proposed by Wang (1997). The order parameter `eta` determines the weight assigned to larger observations. When `eta = 0`, the calculation reduces to the ordinary sample L-moments.

Usage

```
lhmoms(data, eta = NULL, nmom = 5)
```

Arguments

`data`	A numeric vector of data values.
`eta`	A non-negative integer (or a sequence of integers) between 0 and 4 representing the order of the LH-moments.
`nmom`	An integer specifying the maximum number of moments to compute (default is 5).

Value

A list containing:

- `eta`: The order(s) of the LH-moments.
- `lambdas`: A matrix of the calculated sample LH-moments.
- `ratios`: A matrix of the calculated sample LH-moment ratios.

References

Wang, Q. J. (1997). Using higher order L-moments for regional flood frequency analysis. *Water Resources Research*, 33(12), 2841-2848.

`sarakham` *Annual Maximum Daily Temperature for Maha Sarakham, Thailand*

Description

A dataset containing the annual maximum series (AMS) of daily temperature measurements in Maha Sarakham, Thailand, from 1985 to 2025.

Usage

```
data(sarakham)
```

Format

A data frame with 41 rows and 2 variables:

year The observation year (1985-2025).

temperature The annual maximum daily temperature in degrees Celsius (°C).

Source

Thai Meteorological Department (TMD)

Examples

```
data(sarakham)
head(sarakham)
```

`wang.test.lhgev` *Wang's Goodness-of-Fit Test for the Generalized Extreme Value (GEV) Distribution*

Description

This function performs a Goodness-of-Fit (GOF) test for the Generalized Extreme Value (GEV) distribution using LH-moments, based on the methodology proposed by Wang (1998). It calculates a Z-test statistic by comparing the sample LH-kurtosis with the theoretical LH-kurtosis. The test is evaluated across LH-moment orders (`eta`) from 0 to 4.

Usage

```
wang.test.lhgev(data)
```

Arguments

`data`	A numeric vector of data values.

Value

A data frame containing the GOF test results for each `eta` value (from 0 to 4), with the following columns:

- `eta`: The order of the LH-moments.
- `z.test`: The calculated Z-statistic for the GOF test.
- `cond.sigma`: The conditional standard deviation of the sample LH-kurtosis.
- `p.value`: The two-sided p-value corresponding to the Z-test statistic.

References

Wang, Q. J. (1998). Approximate goodness-of-fit tests of fitted generalized extreme value distributions using LH moments. *Water Resources Research*, 34(12), 3497-3502.

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